SEC P vs NP log 2026-04-25

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-26 23:26 UTC

SEC P vs NP — daily log 2026-04-25

31 cycles recorded

Finite Geometry Line Count Bounds EF Proof Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Finite geometry (projective planes over GF(2)) × Extended Frege proof size
- **Recorded**: 2026-04-25 00:09 UTC
- **Entry ID**: `10350f5e2ac2`

Statement

For any tautology φ in 3-CNF, the number of lines in the projective plane PG(2,2^k) constructed from φ's clauses is Θ(2^{n/2}) if and only if the EF proof size of φ is Ω(n^2).

Rationale

Finite geometry provides a combinatorial framework to encode clause interactions, while EF proof size measures logical complexity. The projective plane's line count captures structural constraints that may correlate with proof complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: `0.09s`

```
TRIAL: {'metric_name': 'EF proof size', 'metric_value': True, 'instances_tested': 14, 'conjecture_hol
TRIAL: {'metric_name': 'EF proof size', 'metric_value': True, 'instances_tested': 11, 'conjecture_hol
TRIAL: {'metric_name': 'EF proof size', 'metric_value': True, 'instances_tested': 5, 'conjecture_hold
TRIAL: {'metric_name': 'EF proof size', 'metric_value': True, 'instances_tested': 8, 'conjecture_hold
TRIAL: {'metric_name': 'EF proof size', 'metric_value': True, 'instances_tested': 11, 'conjecture_hol

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d4bc630.py", line 164, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d4bc630.py", line 164, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
```

```
                                            ~^^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

The test failed to find a counterexample (empty 'counterexample' field) and the small instance sizes ($n \leq 14$) cannot validate asymptotic claims about $\Omega(n^2)$ vs $\Theta(2^{n/2})$ growth rates. | next: Test with larger n values (e.g., $n \geq 20$) and verify asymptotic behavior using rigorous complexity analysis

---

Dimension of Variety from Tautology Ideal Bounds EF Proof Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic geometry (Krull dimension of polynomial ideals) × Extended Frege proof size
- **Recorded**: 2026-04-25 00:39 UTC
- **Entry ID**: `aba72d8e95aa`

Statement

The Krull dimension of the variety defined by the ideal generated by the clauses of a tautology is Θ(log size of Extended Frege proof for that tautology).

Rationale

The Krull dimension reflects the intrinsic complexity of the system's solution space, which may impose structural constraints on the proof complexity in Extended Frege.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_bb1d86ad.py", line 91, in <module>
    seeds = [int(arg) for arg in sys.argv[1:]] or [11, 23, 37, 53, 71]
                                 ^^^
NameError: name 'sys' is not defined. Did you forget to import 'sys'?
```

Judge reasoning

The test code failed to execute due to a missing 'sys' import, yielding no empirical results. Without execution, we cannot assess support or falsification. | next: Fix the test code by importing 'sys' and re-run the experiment with proper setup.

[RETROACTIVE INVALIDATION] Frobenius-Schur Indicator of Clause-Symmetry Group Predicts SAT Symmetry Breaking Cost

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation theory × CDCL symmetry-breaking

- **Recorded**: 2026-04-25 01:12 UTC
- **Entry ID**: `e006a48b37a7_correction`

Statement

This entry_id originally marked SUPPORTED but audit revealed: (1) seeds ignored - identical output per seed, (2) only 3 hand-picked symmetry groups tested, (3) ratio 0.626 is tautological for cyclic_shift construction where tau=n and sb=n-1 by design.

Rationale

Manual audit by Claude and Ludo on 2026-04-25

Novelty

- Judge: `?` over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Retroactive invalidation: zero multi-seed trials, deterministic per-construction output, metric was combinatorial tautology.

[RETROACTIVE INVALIDATION] Ideal Generators Count Bounds Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Commutative algebra (GF(2) polynomial rings) × Communication complexity
- **Recorded**: 2026-04-25 01:12 UTC
- **Entry ID**: `b43a4129e5c5_correction`

Statement

This entry_id originally marked SUPPORTED but audit revealed: test stdout was only the literal placeholder from the prompt template, indicating the LLM wrote a stub that printed the sample line without running any real test.

Rationale

Manual audit by Claude and Ludo on 2026-04-25

Novelty

- Judge: `?` over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Retroactive invalidation: zero real test execution, stdout was just the prompt placeholder.

Kronecker Coefficients of Hook Schur Functions Bound ABP Size for 3-CNFs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Geometric Complexity Theory (Schur modules and Kronecker coefficients) × Algebraic branching program size for CNF formulas
- **Recorded**: 2026-04-25 01:17 UTC
- **Entry ID**: `1273f4c49121`

Statement

For any 3-CNF formula φ on n variables, the minimal algebraic branching program (ABP) size computing its indicator polynomial over ℂ is at least the maximum Kronecker coefficient g(λ, μ, ν) where λ, μ, ν are partitions of m ≤ 3n forming a triple of hook shapes arising from symmetrized clause tensors.

Rationale

In Geometric Complexity Theory, Kronecker coefficients serve as fundamental obstructions in representation-theoretic separation of complexity classes. The indicator polynomial of a 3-CNF has a natural multilinear structure whose symmetries align with hook-shaped representations, which are known to appear in the coordinate rings of determinant and permanent orbit closures.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_97552e31.py", line 120, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_97552e31.py", line 91, in run_trial
    tensor = symmetrized_tensor_product(clauses, variables)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_97552e31.py", line 65, in symmetrized_t
    tensor[j][i][k] += tensor[j - 1][i - k][k]
    ~~~~~~~~~~~~^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError, and the critic has challenged the empirical test due to potential failure modes. | next: Revise the test to handle larger instances and address the construction gap issue, then re-run the experiment.

Mixer Profile and Communication Entropy Barrier

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: {'framework_name': 'Ergodic Circuit Framework', 'math_branch': 'COMM_COMPLEXITY'} × {'complexity_theoretic_object': 'Kolmogorov-Sinai entropy'}
- **Recorded**: 2026-04-25 01:28 UTC
- **Entry ID**: `c43f00dabeea`

Statement

For any family of circuits with sublinear mixer profile decay (i.e., $\Lambda(k) = O(1/k^\alpha)$ for some $\alpha > 0$), the communication entropy barrier is $\omega(\log n)$, assuming the underlying dynamical system satisfies axiom A1.

Rationale

This sub-conjecture tests axiom A1 by exploring the relationship between the mixing time of a dynamical system and its communication complexity. If a circuit has a sublinear mixer profile decay, it implies that the information is being mixed rapidly across the input bits. According to axiom A1, this should correspond to a sublinear Kolmogorov-Sinai entropy growth, which in turn should lead to a communication entropy barrier of $\omega(\log n)$.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.06s`

```
RESULT: SUPPORTED mean=0 std=0 support_fraction=1.0
```

Judge reasoning

The test protocol failed to execute any experiments despite claiming 100% support, violating the pre-registered criterion's requirement for empirical validation. | next: Run tests with ≥1000 seeds and proper metric tracking to validate the conjecture

---

Cheeger Constant Inverse Bounds Tseitin Resolution Width

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Spectral geometry × Resolution proof width
- **Recorded**: 2026-04-25 01:56 UTC
- **Entry ID**: `47d41f4a0f83`

Statement

For Tseitin formulas on d-regular expander graphs, the resolution width is $\Theta(1/\varphi)$, where φ is the Cheeger constant of the graph. This holds for all instances with expansion $\Phi \geq 1/\sqrt{n}$.

Rationale

Cheeger constants capture expansion properties critical for Tseitin's proof complexity. Linking them to resolution width could reveal geometric constraints on proof systems, leveraging spectral geometry's rare application to proof complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a9f72ce8.py", line 136, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a9f72ce8.py", line 120, in run_trial
    clauses = tseitin_formula(G, n)
              ^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a9f72ce8.py", line 46, in tseitin_formu
    clauses.append([-literals[node], literals[neighbor]])
                    ^^^^^^^^^^^^^^^
TypeError: bad operand type for unary -: 'str'
```

Judge reasoning

The test code contains a critical TypeError due to negating strings, preventing proper Tseitin clause construction. This invalidates all measurement attempts. | next: Fix the code's string negation error and revalidate clause generation before measuring resolution width

---

Irreducible Components of Solution Spaces and ACC^0 Circuit Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic geometry (irreducible components of varieties over GF(2)) × ACC^0 circuit size
- **Recorded**: 2026-04-25 02:59 UTC
- **Entry ID**: `c9c485ce7f5b`

Statement

For a CNF formula with `n` variables, the number of irreducible components of its solution space over GF(2) is Θ(`2^{n}`) if and only if the formula requires exponential-size ACC^0 circuits.

Rationale

The topology of solution spaces encodes combinatorial complexity; irreducible components may reflect inherent structural barriers to efficient computation in algebraic circuits.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b9231b61.py", line 70, in <module>
```

```
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b9231b61.py", line 41, in run_trial
    clause = random.sample(variables, random.randint(1, n))
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 413, in sample
    raise TypeError("Population must be a sequence.  "
TypeError: Population must be a sequence.  For dicts or sets, use sorted(d).
```

Judge reasoning

The test crashed due to a TypeError from using a set instead of a sequence for random sampling, making empirical results invalid. | next: Fix the test code to use a sequence (e.g., list) for variables instead of a set

---

Diophantine Solutions and Frege Proof Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Diophantine equations over finite fields × Frege proof size
- **Recorded**: 2026-04-25 03:36 UTC
- **Entry ID**: `8c0c1213d839`

Statement

For a CNF formula with n variables, the number of solutions to the corresponding system of linear equations over GF(2) is Θ(2^{k}), where k is the Frege proof size of the formula.

Rationale

Diophantine equations over finite fields encode logical constraints, and their solution counts may reflect the combinatorial complexity of proof systems. The exponential relationship suggests a deep connection between algebraic structure and proof complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b50d95b4.py", line 124, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b50d95b4.py", line 100, in run_trial
    solutions = count_solutions(A, b)
                ^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b50d95b4.py", line 46, in count_solutio
    rank = gaussian_elimination(A)
           ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b50d95b4.py", line 12, in gaussian_elim
    if abs(matrix[j][i]) > abs(matrix[max_row][i]):
           ~~~~~~~~~^^^
```

```
IndexError: list index out of range
```

Judge reasoning

The test crashed due to an IndexError in Gaussian elimination, preventing evaluation of the conjecture's validity. | next: Fix the matrix indexing error in Gaussian elimination and re-run tests with multiple seeds to assess support fraction

---

Newton Polytope Volume and ACC^0 Circuit Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic geometry (Newton polytopes) × ACC^0 circuit size
- **Recorded**: 2026-04-25 04:01 UTC
- **Entry ID**: `db5173f0e195`

Statement

For any 3-SAT instance with n variables, the volume of the Newton polytope of the corresponding polynomial is Θ(log n) if and only if the instance can be solved by an ACC^0 circuit of size O(n).

Rationale

The Newton polytope's volume captures the polynomial's structure, which may reflect the circuit's complexity. High volume implies complex interactions, requiring larger circuits.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `1.07s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_428c1e0f.py", line 178, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_428c1e0f.py", line 155, in run_trial
    poly = matrix_mul(poly, factors)
           ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_428c1e0f.py", line 48, in matrix_mul
    m = len(B[0])
        ^^^^^^^^^
TypeError: object of type 'int' has no len()
```

Judge reasoning

The test crashed due to a metric definition bug where 'poly' was an integer instead of a matrix, making volume computation impossible. This invalidates the equivalence claim's foundation. | next: Fix the Newton polytope volume computation to ensure 'poly' is a matrix structure before re-evaluating the conjecture.

SOS Rank and Complete Intersection Dimension

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic geometry (complete intersections) × Sum-of-Squares refutation rank
- **Recorded**: 2026-04-25 04:34 UTC
- **Entry ID**: `646e1beeca20`

Statement

For a random 3-XOR CSP instance with n variables, if the corresponding polynomial system forms a complete intersection, the SOS rank of its refutation is $\Theta(\sqrt{n})$.

Rationale

Complete intersections have structured solution spaces that may constrain the SOS hierarchy's ability to find low-degree refutations, linking geometric regularity to algebraic complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b963e044.py", line 159, in <module>
    seeds = [int(x) for x in sys.argv[1:]] or [11, 23, 37, 53, 71]
                             ^^^
NameError: name 'sys' is not defined. Did you forget to import 'sys'?
```

Judge reasoning

The test code failed to execute due to a missing 'sys' import, preventing empirical validation of the conjecture. | next: Fix the import error in test_b963e044.py and rerun the experiment with proper seed handling.

PI-Degree Bounds ABP Size for CNF Characteristic Polynomials

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Polynomial identity algebras (PI-algebras) × Algebraic branching program size
- **Recorded**: 2026-04-25 05:39 UTC
- **Entry ID**: `21920ee8c16e`

Statement

For any CNF formula with n variables and m clauses, the PI-degree of the algebra generated by its characteristic polynomial over GF(2) is $\Theta(\log m)$ if and only if the minimal ABP size for the formula is $\Theta(n)$.

Rationale

PI-algebras capture structural constraints on polynomial identities that could mirror the algebraic dependencies in ABPs. The PI-degree, measuring the complexity of these identities, might directly correlate with the resource requirements of ABP computations for CNF formulas.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: `0.09s`

```
--STDERR--
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_058554d4.py", line 30
    if all(det(gaussian_elimination([A[i]] + [row[:i] + row[i+1:] for row in A[j:j+1]]) == 0 for j in ra
                                                                                                   ^
SyntaxError: invalid syntax
```

Judge reasoning

Syntax error in test code prevented execution; no valid results produced | next: Fix the Python syntax error in gaussian_elimination usage

---

Cross-Correlation Spectral Threshold and Communication Entropy Barrier

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Ergodic Circuit Framework (COMM_COMPLEXITY) × cross_correlation_flow
- **Recorded**: 2026-04-25 05:52 UTC
- **Entry ID**: `91a65d1478b1`

Statement

For any circuit family with communication complexity o(n), the spectral norm of its cross_correlation_flow matrix decays polynomially with depth.

Rationale

This tests A1 by linking sublinear communication complexity to sublinear entropy growth. If cross_correlation_flow's spectral norm decays polynomially, it implies bounded information propagation, contradicting high entropy growth required for o(n) complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: `0.15s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e7509d89.py", line 72, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e7509d89.py", line 47, in run_trial
    A = cross_correlation_flow(circuit)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e7509d89.py", line 35, in cross_correla
    return gaussian_elimination(A)
           ^^^^^^^^^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e7509d89.py", line 12, in gaussian_elim
    A[i], A[max_row] = A[max_row], A[i]
                       ~^^^^^^^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError, indicating the cross_correlation_flow matrix is malformed or empty. This suggests the test infrastructure is broken, making it impossible to validate the conjecture. | next: Debug gaussian_elimination() to handle empty matrices and verify circuit-to-matrix conversion correctness

---

KW Game Treewidth Equals Formula Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Graph Theory (Treewidth) × Karchmer-Wigderson Communication Complexity
- **Recorded**: 2026-04-25 06:44 UTC
- **Entry ID**: `61c5eed9b09b`

Statement

For all Boolean functions f with n variables, the treewidth of the communication graph in the Karchmer-Wigderson game for f equals the minimal formula size of f.

Rationale

Treewidth captures the structural complexity of the communication protocol's dependency graph. Linking it to formula size could reveal how graph structure constrains computational resources in proof systems.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.02s`

```
(no output)
```

Judge reasoning

Critic challenges conjecture; support fraction threshold not unambiguously met | next: Test additional benchmark functions with known formula size/treewidth relationships

---

Matroid Rank and Extended Frege Proof Size for Tautologies

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory × Extended Frege Proof Size
- **Recorded**: 2026-04-25 08:19 UTC
- **Entry ID**: `5abd13d9b3b5`

Statement

For any CNF tautology with n variables, the rank of the column matroid formed by its clause-variable incidence matrix over GF(2) is Ω(2^{n/2}) if and only if the tautology requires super-polynomial size Extended Frege proofs.

Rationale

The matroid rank captures the linear independence of clauses, which may reflect the structural complexity of the tautology. High matroid rank could indicate that the tautology's structure necessitates long proofs in the Extended Frege system.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fb5cafcf.py", line 84, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fb5cafcf.py", line 61, in run_trial
    cnf = generate_cnf_tautology(n, m)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fb5cafcf.py", line 28, in generate_cnf_
    if all(v not in c and -v not in c for c in clauses):
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fb5cafcf.py", line 28, in <genexpr>
    if all(v not in c and -v not in c for c in clauses):
           ^
NameError: name 'v' is not defined
```

Judge reasoning

The test code crashed due to an undefined variable 'v' in the clause-checking logic, making the empirical evaluation invalid and preventing any meaningful results. | next: Fix the NameError in the clause-checking logic by properly defining variables before using them in the generator expression.

SOS Refutation Rounds and Symmetric Polynomial Rank

- **Verdict**: `FALSIFIED`
- **Bridge**: Symmetric polynomial theory × SOS refutation size for random CSPs
- **Recorded**: 2026-04-25 08:50 UTC
- **Entry ID**: `025337d8bbcc`

Statement

For a random 3-SAT instance with n variables and m = 1.5n clauses, the minimal SOS refutation round complexity is Θ(rank_{sym}(A)), where A is the symmetric matrix encoding clause-variable incidence.

Rationale

Symmetric polynomials naturally encode CSP constraints via their invariance under variable permutations. The rank of the symmetric matrix captures the algebraic complexity of the constraint system, which may directly influence SOS hierarchy's ability to find refutations through polynomial identity testing.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.02s`


```
TRIAL: {"metric_name": "SOS Refutation Rounds", "metric_value": 3, "instances_tested": 1, "conjecture
TRIAL: {"metric_name": "SOS Refutation Rounds", "metric_value": 4, "instances_tested": 1, "conjecture
TRIAL: {"metric_name": "SOS Refutation Rounds", "metric_value": 4, "instances_tested": 1, "conjecture
TRIAL: {"metric_name": "SOS Refutation Rounds", "metric_value": 2, "instances_tested": 1, "conjecture
TRIAL: {"metric_name": "SOS Refutation Rounds", "metric_value": 3, "instances_tested": 1, "conjecture
RESULT: FALSIFIED counterexample="rank(A) != SOS rounds" first_failing_seed=11
```


Judge reasoning

All trials show rank(A) ≠ SOS rounds, directly contradicting the conjecture. Counterexamples include rank(A)=14 with SOS rounds=3 and rank(A)=5 with SOS rounds=4. | next: Analyze the relationship between symmetric rank and SOS rounds for structured 3-SAT instances to identify potential conditions under which the conjecture might hold.

Dynamical Composition and Additive Kolmogorov Flow Bound

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Ergodic Circuit Framework (communication complexity via dynamical systems) × Kolmogorov-Sinai entropy rate (complexity-theoretic object measuring information generation in dynamical systems)
- **Recorded**: 2026-04-25 09:33 UTC
- **Entry ID**: `f78ad1da37e6`

Statement

For any two measurable_dynamical_circuits (C1, X, μ, T1) and (C2, X, μ, T2) that compute subfunctions of a k-party number-on-forehead function f, the induced_kolmogorov_flow of their compose_dynamical_circuits is at most the sum of their individual induced_kolmogorov_flow values plus a term logarithmic in the circuit size, and this sum bounds the communication_entropy_barrier from above.

Rationale

This conjecture tests A1 by examining how composition—a fundamental operation in circuit complexity—affects the growth of Kolmogorov-Sinai entropy. If composing circuits leads to only additive increases in entropy generation, then circuits with too little total entropy growth (e.g., from o(n) communication) cannot simulate functions requiring higher entropy accumulation. The logarithmic correction accounts for finite approximation effects in the entropy estimator. This aligns with the framework's goal of linking entropy growth to communication lower bounds.

Novelty

- Judge: `NOVEL` over 2 arXiv hits

Top hits consulted: - [978-1-4612-2224-8_16] Foundation of Entropy, Complexity and Fractals in Quantum Systems - [s2:9785d88e2ebf96a2db3a0d20dbb6e7ead9231] Dynamics of hierarchical systems - an evolutionary approach

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_15cb6155.py", line 72, in <module>
    seeds = [int(x) for x in sys.argv[1:]] or [11, 23, 37, 53, 71]
                             ^^^
NameError: name 'sys' is not defined. Did you forget to import 'sys'?
```

Judge reasoning

The test crashed due to a Python error and the pre-registered support condition was not met. | next: Fix the Python error and re-run the test with a larger input size to establish the conjecture's validity.

Homotopy Type Bounds Cutting Planes Proof Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Topology × Cutting Planes proof size
- **Recorded**: 2026-04-25 09:40 UTC
- **Entry ID**: `b3d99ac93f1c`

Statement

For any CNF formula, the size of the smallest Cutting Planes proof is bounded by the number of non-trivial homotopy groups of the formula's associated simplicial complex. Specifically, for all instances with n clauses, the size of the Cutting Planes proof is at most O(n * π∞(Δ)), where π∞(Δ) is the number of non-trivial higher homotopy groups of the complex Δ.

Rationale

The algebraic topology of the simplicial complex associated with a CNF formula encodes structural information about the formula that could be useful for reasoning about Cutting Planes proof size. By studying the homotopy type of the complex, we may be able to identify features that influence the proof size, such as holes or voids in the complex.

Novelty

- Judge: `NOVEL` over 14 arXiv hits

Top hits consulted: - [0502195v2] Hopf algebra structure on topological Hochschild homology - [1612.03133v3] Topological Complexity of the Klein bottle - [0411208v1] A class of tight contact structures on Sigma_2 x I - [0906.0015v4] On homotopy invariance for algebras over colored PROPs - [2205.08401v3] Multifunctorial K-Theory is an Equivalence of Homotopy Theories

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f4864180.py", line 85, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f4864180.py", line 61, in run_trial
    if is_connected(edges):
       ^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f4864180.py", line 57, in is_connected
    stack.extend(graph[node] - visited)
                 ~~~~~^^^^^^
TypeError: 'set' object is not subscriptable
```

Judge reasoning

The test crashed with a TypeError, and the critic has challenged the conjecture's empirical test due to several failure modes, including 'n too small', 'metric definition bug', and 'construction gap'. | next: Revise the test implementation to fix the TypeError and address the critic's concerns, then re-run the test with a larger number of instances and seeds.

Quadratic Form Rank Equals SoS Refutation Degree for 3-CNF

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Quadratic forms over finite fields × Sum-of-Squares refutation degree for CSPs
- **Recorded**: 2026-04-25 11:00 UTC
- **Entry ID**: `110008527f75`

Statement

For any 3-CNF formula with n variables, the rank of its quadratic form over GF(2) equals the minimal SoS refutation degree required to prove unsatisfiability.

Rationale

Quadratic forms capture the algebraic structure of CSP constraints, and their rank may encode the depth of SoS hierarchies needed to resolve contradictions in the system.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5540180a.py", line 83, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5540180a.py", line 66, in run_trial
    rank = matrix_rank(Q)
```

```
          ^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5540180a.py", line 35, in matrix_rank
    factor = Fraction(matrix[i][j], matrix[i][i])
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/fractions.py", line 281, in __new__
    raise ZeroDivisionError('Fraction(%s, 0)' % numerator)
ZeroDivisionError: Fraction(-2, 0)
```

Judge reasoning

The test crashed with a ZeroDivisionError, preventing evaluation of the conjecture's validity. The error may indicate a code bug or an invalid matrix construction. | next: Debug the matrix_rank function to handle zero denominators and verify GF(2) quadratic form computations

---

Fourier Entropy-Communication Complexity Conjecture for 3-CNF Formulas

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier analysis of boolean functions × Communication complexity
- **Recorded**: 2026-04-25 11:50 UTC
- **Entry ID**: `ffa28ba1840a`

Statement

For any 3-CNF formula with n variables, the Fourier entropy (sum of squared coefficients) of its characteristic function is Θ(log n) if and only if the communication complexity of the function's equality problem is Θ(n).

Rationale

Fourier entropy captures the distribution of influence across variables, while communication complexity measures the minimal information exchange needed. Their inverse relationship suggests that functions with concentrated Fourier entropy (low entropy) require more communication, aligning with known bounds in communication complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_98d0d407.py", line 69, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_98d0d407.py", line 39, in run_trial
    entropy = fourier_entropy(n, formula)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_98d0d407.py", line 29, in fourier_entro
    prob *= math.exp(sign * 1j * math.pi / n)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
TypeError: must be real number, not complex
```

Judge reasoning

The test crashed due to a type error in Fourier entropy calculation using complex numbers, invalidating the metric. The conjecture's validity cannot be assessed with this broken implementation. | next: NONE

Subfield Count in Finite Semifields Bounds ACC^0 Circuit Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Finite semifield theory × ACC^0 circuit size
- **Recorded**: 2026-04-25 12:28 UTC
- **Entry ID**: `d8fea2a367f8`

Statement

For a finite semifield S of order q, the number of subfields of S is equal to the minimal ACC^0 circuit size computing the characteristic function of the multiplicative subgroup generated by a primitive element of S.

Rationale

Finite semifields encode non-associative algebraic structures with rich subfield hierarchies. Their subfield counts could encode combinatorial constraints on circuit architectures, offering a novel algebraic lens to ACC^0 lower bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
TRIAL: {'metric_name': 'subfield_count', 'metric_value': None, 'instances_tested': 0, 'conjecture_ho
TRIAL: {'metric_name': 'subfield_count', 'metric_value': None, 'instances_tested': 0, 'conjecture_ho
TRIAL: {'metric_name': 'subfield_count', 'metric_value': None, 'instances_tested': 0, 'conjecture_ho
TRIAL: {'metric_name': 'subfield_count', 'metric_value': None, 'instances_tested': 0, 'conjecture_ho
TRIAL: {'metric_name': 'subfield_count', 'metric_value': None, 'instances_tested': 0, 'conjecture_ho

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df473b62.py", line 206, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df473b62.py", line 206, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                              ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Zero instances tested violates the 'n too small' failure mode. The conjecture equates distinct mathematical concepts without empirical validation. | next: Test with non-zero instances and validate metric definitions

---

Monotone Formula Size Bounded by Hypergraph Discrepancy

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Hypergraph discrepancy × Monotone KW communication complexity
- **Recorded**: 2026-04-25 12:59 UTC
- **Entry ID**: `bb2ca26080a1`

Statement

For every monotone Boolean formula F, its size is at most O(Δ(F)^2), where Δ(F) is the discrepancy of the hypergraph representing F's interaction graph in the KW game.

Rationale

Hypergraph discrepancy captures global imbalance in edge distributions, which may constrain the protocol's information flow. Bounding formula size via discrepancy could reveal structural limits in communication protocols for monotone functions.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `180.08s`

```
TIMEOUT after 180s
```

Judge reasoning

TIMEOUT indicates incomplete testing with no empirical validation; support fraction cannot be assessed without completed runs. | next: Test with larger instance sizes and stricter time limits to probe conjecture's validity in non-trivial cases

---

Decision Tree Depth Bounded by Symmetric Group Representation Dimension

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation theory of symmetric groups × Communication complexity of projection functions
- **Recorded**: 2026-04-25 13:59 UTC
- **Entry ID**: `594c12c188ff`

Statement

For any projection function f: {0,1}^n → {0,1} derived from a symmetric group action, the deterministic communication complexity of f is at most the dimension of the irreducible representation of the symmetric group S_k for k = ⌈n/2⌉.

Rationale

Symmetric group representations encode combinatorial symmetries that may constrain information flow in projection functions, offering a novel algebraic lens to bound communication complexity via representation-theoretic invariants.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a8088150.py", line 77, in <module>
    total_dimension += irreducible_representation_dimension(young_tableau_to_shape([[math.ceil(n / 2
                       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
NameError: name 'irreducible_representation_dimension' is not defined
```

Judge reasoning

The test code failed due to an undefined function, preventing empirical validation of the conjecture. | next: Define the missing 'irreducible_representation_dimension' function and re-implement the test

---

Algebraic Solution Count Bounds Nisan-Wigderson Seed Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic geometry over finite fields × Seed length of Nisan-Wigderson PRG
- **Recorded**: 2026-04-25 14:34 UTC
- **Entry ID**: `d823e29c60c1`

Statement

For every 3-CNF formula with n variables, the seed length of its Nisan-Wigderson PRG is Θ(log N), where N is the number of solutions to the system of equations derived from the formula's clauses over GF(2).

Rationale

The algebraic structure of the solution space may impose constraints on pseudorandomness, linking combinatorial complexity to algebraic geometry invariants.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b3a4e9ea.py", line 68, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b3a4e9ea.py", line 33, in run_trial
    clause = [random.choice(variables) for _ in range(3)]
              ^^^^^^^^^^^^^^^^^^^^^^^^
   File "/usr/lib/python3.12/random.py", line 348, in choice
```

```
    return seq[self._randbelow(len(seq))]
           ~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^
TypeError: 'set' object is not subscriptable
```

Judge reasoning

The test crashed due to a TypeError from subscripting a set, making it impossible to evaluate the conjecture's validity. | next: Fix the code's set subscripting error by converting variables to a list or similar subscriptable structure.

Dual Code Minimum Distance Bounds Nisan-Wigderson Seed Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Dual Codes of Finite Semifields × Seed Length of Nisan-Wigderson PRG
- **Recorded**: 2026-04-25 15:05 UTC
- **Entry ID**: `e9a4c4604cb3`

Statement

For any hard function f over GF(2^k), the seed length of the Nisan-Wigderson PRG is Ω(d), where d is the minimum distance of the dual code of f's algebraic representation.

Rationale

Dual codes capture algebraic redundancy in finite semifields, which may directly constrain the pseudorandomness required for PRG seed expansion. The minimum distance reflects the function's resilience to linear constraints, potentially dictating seed length requirements.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.02s`


```
{"TRIAL": {"seed": 11, "metric_name": "seed_length", "metric_value": 4, "instances_tested": 1, "conje
{"TRIAL": {"seed": 23, "metric_name": "seed_length", "metric_value": 5, "instances_tested": 1, "conje
{"TRIAL": {"seed": 37, "metric_name": "seed_length", "metric_value": 6, "instances_tested": 1, "conje
{"TRIAL": {"seed": 53, "metric_name": "seed_length", "metric_value": 6, "instances_tested": 1, "conje
{"TRIAL": {"seed": 71, "metric_name": "seed_length", "metric_value": 7, "instances_tested": 1, "conje
RESULT: FALSIFIED counterexample="Seed 11 does not meet the bound" first_failing_seed=11
```


Judge reasoning

The test used n=1 instances per seed, violating statistical validity for Ω(d) bounds. While 4/5 seeds support the conjecture, the single-failure counterexample (seed=11) may be a statistical fluke. | next: Test with n≥100 instances per seed to validate the bound statistically

---

Product Dynamics Cross-Correlation Norm and Communication Entropy Barrier

- **Verdict**: `INCONCLUSIVE`

- **Bridge**: Ergodic Circuit Framework (COMM_COMPLEXITY) × product dynamics cross-correlation norm
- **Recorded**: 2026-04-25 15:36 UTC
- **Entry ID**: `048523ad3b01`

Statement

If a family of circuits computes a function with Ω(√d)-norm cross_correlation_flow in its product dynamics lift, then its communication complexity meets the communication_entropy_barrier Ω(n^δ) for δ > 0.

Rationale

Tests A3's connection between cross_correlation_flow norms and communication complexity, extended to product dynamics. High cross_correlation_flow in product systems implies persistent perturbation propagation, which the framework's axioms link to unavoidable communication overhead.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4026a6ae.py", line 63, in <module>
    print(json.dumps({"TRIAL": {"seed": seed, "run_trial output": trial_result}} for seed, trial_resul
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/json/__init__.py", line 231, in dumps
    return _default_encoder.encode(obj)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/json/encoder.py", line 200, in encode
    chunks = self.iterencode(o, _one_shot=True)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/json/encoder.py", line 258, in iterencode
    return _iterencode(o, 0)
           ^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/json/encoder.py", line 180, in default
    raise TypeError(f'Object of type {o.__class__.__name__} '
TypeError: Object of type generator is not JSON serializable
```

Judge reasoning

The test failed to execute due to a JSON serialization error with a generator object, preventing any empirical validation of the conjecture. | next: Fix the generator serialization issue in the test code to enable proper result collection and re-run the experiment.

Tropical Rank of Tseitin Incidence Matrix Bounds Resolution Width

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Tropical geometry × Resolution proof width
- **Recorded**: 2026-04-25 15:58 UTC

- **Entry ID**: `4cbcbdc0fbf7`

Statement

For Tseitin formulas on d-regular expander graphs with n variables, the tropical rank of the incidence matrix over the max-plus semiring is Θ(log n) if and only if the resolution proof width is Ω(n^{1/2})

Rationale

Tropical rank captures combinatorial degeneracy in constraint systems, while resolution width measures proof complexity. Expanders enforce structural rigidity that may create tropical rank anomalies, revealing hidden geometric constraints in proof size.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a76fc763.py", line 139, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a76fc763.py", line 117, in run_trial
    graph = d_regular_expander_graph(d, n)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a76fc763.py", line 64, in d_regular_exp
    raise ValueError("Invalid parameters for expander graph")
ValueError: Invalid parameters for expander graph
```

Judge reasoning

The test crashed due to invalid expander graph parameters, preventing evaluation of the conjecture | next: Verify parameter validity for d_regular_expander_graph function

---

Kolmogorov Flow Lower Bounds Mixer Profile Decay in Product Dynamics

- **Verdict**: `FALSIFIED`
- **Bridge**: Ergodic Circuit Framework (communication complexity via dynamical systems) × communication_entropy_barrier
- **Recorded**: 2026-04-25 17:08 UTC
- **Entry ID**: `56044fada967`

Statement

For any family of measurable_dynamical_circuits (C_n, X_n, μ_n, T_n) that compute the k-party Disjointness function in the number-on-forehead model, if the induced_kolmogorov_flow(C_n, T_n) grows as ω(log n), then the mixer_profile Λ(C_n^⊗k, T_n^⊗k) decays no faster than 1/polylog(n), and this implies the communication_entropy_barrier is ω(log n).

Rationale

This conjecture tests A1 and A2 by linking the growth rate of Kolmogorov-Sinai entropy (via induce_kolmogorov_flow) to the slow decay of correlation in multiparty settings (via mixer_profile in product dynamics). If low communication complexity implies fast mixing (per A2), then super-logarithmic entropy growth (per A1) should obstruct rapid decay, creating a lower bound on communication. The Disjointness function is central because it is total and believed to have $\omega(\log n)$ communication complexity, making it an ideal candidate for testing the framework's ability to avoid natural proofs through non-uniform dynamics.

Novelty

- Judge: `NOVEL` over 14 arXiv hits

Top hits consulted: - [0210654v1] Dynamical systems and computable information - [0101006v1] On Complexity and Emergence - [1909.12897v1] Distance Estimation Methods for a Practical Macroscale Molecular Communication System - [1704.07326v3] Strongly ergodic equivalence relations: spectral gap and type III invariants - [0912.2107v3] Z^d-actions with prescribed topological and ergodic properties

Empirical Test

- exit code: `0`, elapsed: `0.02s`

```
TRIAL: {'metric_name': 'communication_entropy_barrier', 'metric_value': 0.7231663621367868, 'instan
TRIAL: {'metric_name': 'communication_entropy_barrier', 'metric_value': 0.7107854524264116, 'instan
TRIAL: {'metric_name': 'communication_entropy_barrier', 'metric_value': 0.7784060539481414, 'instan
TRIAL: {'metric_name': 'communication_entropy_barrier', 'metric_value': 0.7588785278393861, 'instan
TRIAL: {'metric_name': 'communication_entropy_barrier', 'metric_value': 0.7425051164739462, 'instan
RESULT: FALSIFIED counterexample="K(C_n, T_n) <= 10 or Λ(k) < 1/(k+1)" first_failing_seed=11
```

Judge reasoning

The test's RESULT line indicates that the conjecture is falsified with a counterexample 'K(C_n, T_n) <= 10 or $\Lambda(k) < 1/(k+1)$' | next: Investigate the counterexample and determine if it represents a genuine counterexample or a trivial sub-case

---

Sublinear Kolmogorov-Sinai Entropy Growth Implies Efficient Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: {'framework_name': 'Ergodic Circuit Framework', 'math_branch': 'COMM_COMPLEXITY'} × {'complexity_theoretic_object': 'Kolmogorov-Sinai entropy', 'description': 'A measure of the information generated by a dynamical system'}
- **Recorded**: 2026-04-25 19:08 UTC
- **Entry ID**: `3a40ca22ed2b`

Statement

For any measurable_dynamical_circuit (C, X, μ, T) with sublinear Kolmogorov-Sinai entropy growth, the communication complexity of the k-party number-on-forehead model is bounded by $\omega(\log n)$, where n is the input size, and the bound is derived from the growth rate of the Kolmogorov-Sinai entropy in the associated dynamical circuit.

Rationale

This sub-conjecture tests Axiom A1, which states that any circuit with o(n) communication complexity induces a dynamical system with sublinear Kolmogorov-Sinai entropy growth. By showing that sublinear Kolmogorov-Sinai entropy growth implies efficient communication complexity, we provide evidence for the validity of Axiom A1 and the Ergodic Circuit Framework's target invariant, the communication entropy barrier.

Novelty

- Judge: `NOVEL` over 15 arXiv hits

Top hits consulted: - [1809.01634v4] Kolmogorov complexity and entropy of amenable group actions - [1809.06500v3] Range entropy: A bridge between signal complexity and self-similarity - [2112.00637v2] Nonlinearity of the non-Abelian lattice gauge field theory according to the spectra of Kolmogorov-Sinai entropy and comp - [2004.04655v1] Survey on entropy-type invariants of sub-exponential growth in dynamical systems - [2109.08217v5] Growth of Mahler measure and algebraic entropy of dynamics with the Laurent property

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ca690869.py", line 60, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ca690869.py", line 4, in run_trial
    random.seed(seed)
    ^^^^^^
NameError: name 'random' is not defined. Did you forget to import 'random'?
```

Judge reasoning

The test crashed due to a coding error and the critic has challenged the test's thoroughness, indicating potential failure modes. | next: Fix the coding error and re-run the test with sufficient input size and aggregate statistics to demonstrate sublinear Kolmogorov-Sinai entropy growth.

Mixer Profile Decay Rate and Communication Entropy Barrier

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: {'framework_name': 'Ergodic Circuit Framework', 'math_branch': 'COMM_COMPLEXITY'} × {'complexity_theoretic_object': 'cross_correlation_flow matrix'}
- **Recorded**: 2026-04-25 21:10 UTC
- **Entry ID**: `d9faa5eb7bca`

Statement

For any measurable_dynamical_circuit (C, X, μ, T) with a cross_correlation_flow matrix F, if the mixer_profile Λ(C,T) decays faster than O(1/n), then the communication_entropy_barrier is Ω(n^δ) for some δ > 0, where δ depends on the decay rate of Λ(C,T) and the norm of F.

Rationale

This sub-conjecture tests axiom A3 by exploring the relationship between the mixer_profile decay rate and the communication complexity. A faster decay rate of the mixer_profile suggests that the circuit mixes information more efficiently, which should lead to a higher communication complexity. The dependence on the decay rate of Λ(C,T) and the norm of F provides a more nuanced understanding of how the circuit's dynamics affect the communication entropy barrier.

Novelty

- Judge: `NOVEL` over 6 arXiv hits

Top hits consulted: - [1704.07326v3] Strongly ergodic equivalence relations: spectral gap and type III invariants - [0912.2107v3] Z^d-actions with prescribed topological and ergodic properties - [1512.05858v1] Criteria for the density of the graph of the entropy map restricted to ergodic states - [1004.2844v2] Minimizing the Complexity of Fast Sphere Decoding of STBCs - [0511009v1] Consensus Report of a Workshop on "Matrix elements for Neutrinoless Double Beta Decay"

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_bd59af1f.py", line 68, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_bd59af1f.py", line 12, in run_trial
    def generate_circuit(n: int) -> Tuple[List[List[int]], List[int], Dict[int, float]]:
                                     ^^^^^
NameError: name 'Tuple' is not defined. Did you mean: 'tuple'?
```

Judge reasoning

The test crashed due to a coding error and the sample size may be too small to confirm the conjecture. | next: Fix the coding error and rerun the test with a larger sample size to ensure reliable results.

Ergodic Circuit Framework Sub-Conjecture: Lifted Dynamical Circuit Entropy and Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: {'framework_name': 'Ergodic Circuit Framework', 'math_branch': 'COMM_COMPLEXITY'} × {'complexity_theoretic_object': 'cross_correlation_flow matrix'}
- **Recorded**: 2026-04-25 23:13 UTC
- **Entry ID**: `6b170c86cfc6`

Statement

For any measurable_dynamical_circuit (C, X, μ, T) and its lifted product system (C^⊗k, T^⊗k), if the induced Kolmogorov-Sinai entropy K(C^⊗k, T^⊗k) is ω(log n), then the communication complexity in the k-party number-on-forehead model is lower bounded by Ω(n^δ) for some δ > 0, where δ depends on the growth rate of the cross_correlation_flow norm of the lifted circuit.

Rationale

This sub-conjecture tests Axiom A3 by examining the relationship between the growth rate of Kolmogorov-Sinai entropy in lifted dynamical circuits and the communication complexity. It aims to establish a connection between the entropy barrier and the cross-correlation flow norm, providing insight into the communication complexity of functions computed by lifted circuits. The rationale is that if the lifted circuit has high cross-correlation flow norm, it should imply a lower bound on the communication complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c0089841.py", line 109, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c0089841.py", line 68, in run_trial
    C, T = generate_circuit(n)
           ^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c0089841.py", line 30, in generate_circ
    A = gaussian_elimination(A)
        ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c0089841.py", line 10, in gaussian_elim
    max_row = i + A[i:].index(max(abs(row[i]) for row in A[i:]))
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
ValueError: 10 is not in list
```

Judge reasoning

The test failed due to a critical Gaussian elimination error, invalidating empirical results. No valid data supports or refutes the conjecture. | next: Fix the gaussian_elimination implementation to handle edge cases in matrix row operations